

Auto Loans - Preliminary Model

Code and presentation completed in a weekend

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Executive Summary

Candidate Models & Performance

Model A
High-lag model

71%
Adj. R²

Model B
Medium-lag model
Proposed

69%
Adj. R²

Model C
No LWVS variable

57%
Adj. R²

Model Selection Process

- 01** Profile the data
- 02** Variable Transformation
- 03** Standardization
- 04** Stationarity Testing
- 05** Check for Intuitive Sign
- 06** Variable selection
- 07** Testing and Visual inspection

Variables vs. Net Charge Off Rate

+ POSITIVE CORRELATION WITH DEFAULTS

Unemployment

Should correlate **positively** with default rates as job loss directly reduces a borrower's ability to repay loans.

High CPI (Inflation)

Indicative of challenging economic conditions for consumers. Increased living costs strain household budgets, correlating **positively** with high default rates.

MUVVI

Generally correlates with CPI for used cars. Rising used vehicle prices may put additional pressure on buyers, leading to a **positive** correlation.

- NEGATIVE & MIXED RELATIONSHIPS

High GDP Growth

Strong economic expansion reflects a healthy business climate, which should correlate **negatively** with borrower defaults.

High HPI Growth

Higher home values signal stronger asset backing and economic health, correlating **negatively** with default rates.

RDPI (Disposable Income)

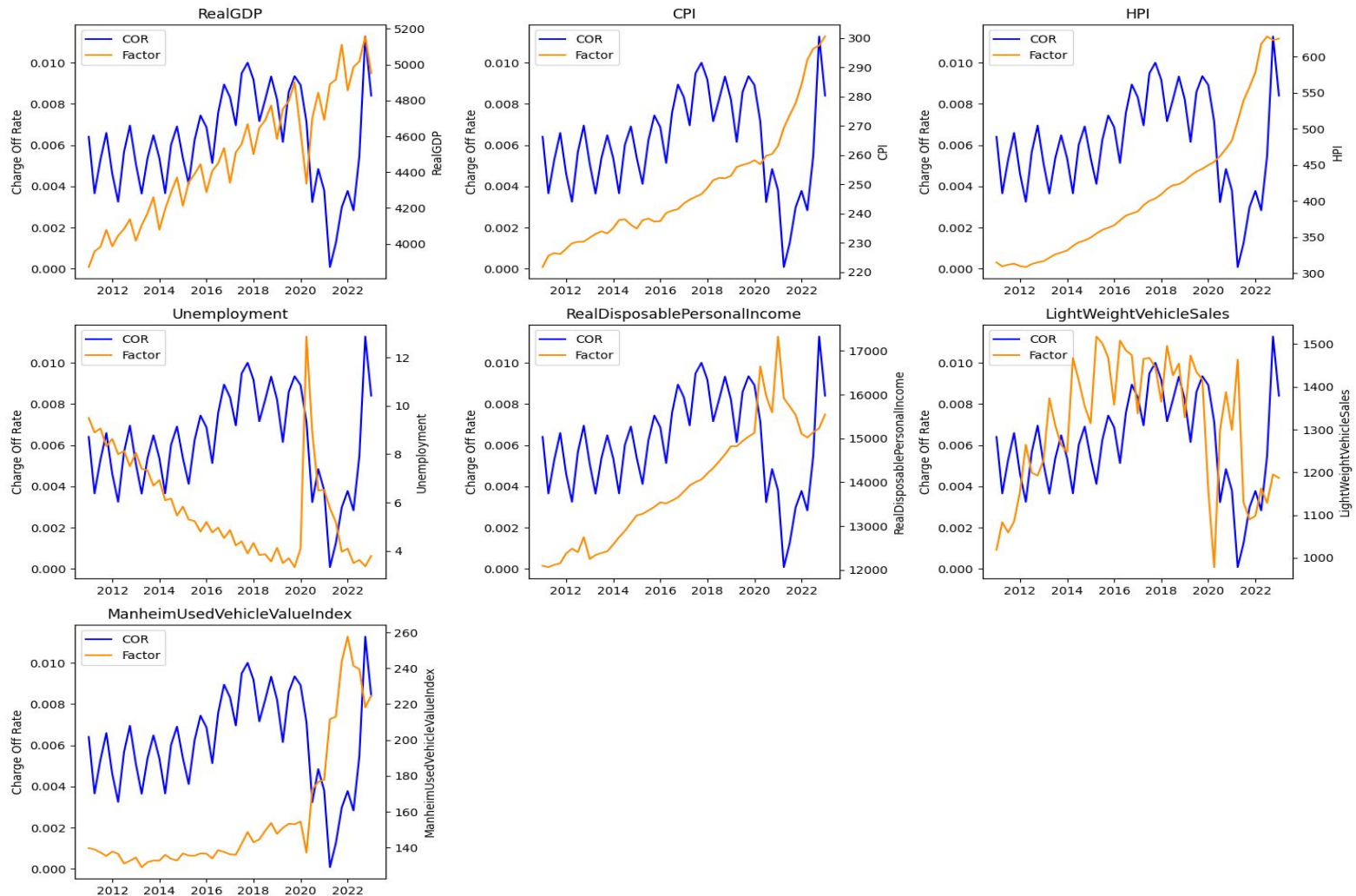
Indicates more liquid cash in hand for customers, which inherently **negatively** correlates with defaults.

LWVS (Vehicle Sales)

Generally matches GDP growth (expected negative correlation). However, historically, higher auto sales surges have occasionally **correlated positively** with default rates.

Data Profiling

★ Key Takeaway: Many variables are non-stationary, requiring transforms



Variable Transformation

Base Transforms

YoY — Year over Year

QoQ — Quarter over Quarter

dQ — 1 Quarter Difference

ldQ — Log 1 Quarter Difference

lvl — Level (Raw Variable Value)

Parameters & Constraints

Moving Average: 1 - 3 Quarters

Lags: 0 - 4 Quarters

Max Delay Limit: 8 Quarters

(lag + moving average lag + transform lag ≤ 8)

432 Total Variable Transforms Considered

Standardization, Stationarity & Intuitive Sign

Standardization

Variables are standardized to ensure zero mean and unit variance before analysis

$$z = \frac{x - \mu}{\sigma}$$

μ = Mean

σ = Standard Deviation

Stationarity

Augmented Dickey-Fuller (ADF) test is applied to check for stationarity.

115

Variable transforms found to be stationary

Intuitive Sign

Relationship to target variable based on univariate regression:

Real GDP (-)

Unemployment (+)

HPI (-) / **CPI** (+)

Real Disposable Income (-)

Vehicle Sales (?)

Manheim Index (+)

62

Variable transforms have intuitive sign

Model Selection

Model Search Space

Tested **72,339** variable transform combinations, with strict constraints:

- 2 to 4 variables per model
- No more than 1 of each root variable
- Modeling data: 80/20 train/test split

Selection Criteria

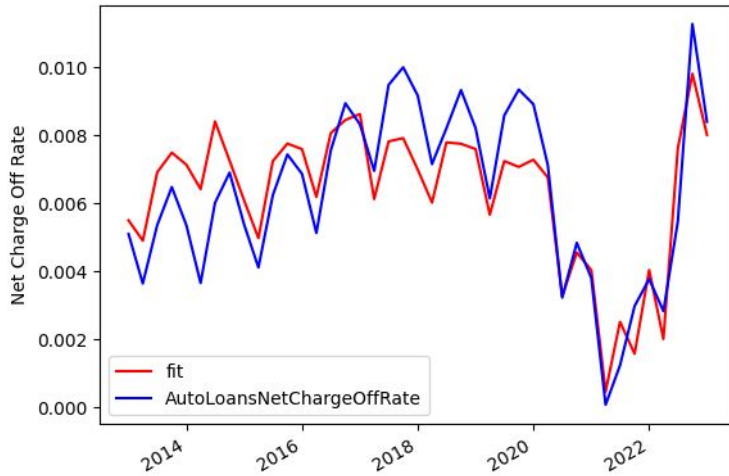
All passing models must meet these thresholds:

- **P-values** < 5%
- **Intuitive** coefficient signs
- **VIF** < 4
- **Adjusted R-squared** > 40%

928 passing models corresponding to **7** root variable combinations:

f1		f2		f3		f4	f5	R2_adj_full
RealGDP_yoy_I0_m1		HPI_ldq_I0_m1		RealDisposablePersonalIncome_yoy_I2_m2		LightWeightVehicleSales_lvl_I1_m1	const	0.714552
HPI_ldq_I0_m1		RealDisposablePersonalIncome_yoy_I2_m2		LightWeightVehicleSales_lvl_I1_m1		const	NaN	0.692765
RealGDP_ldq_I2_m3		RealDisposablePersonalIncome_yoy_I3_m1		LightWeightVehicleSales_lvl_I2_m1		const	NaN	0.640629
HPI_ldq_I0_m1		LightWeightVehicleSales_dq_I0_m1		const		NaN	NaN	0.579227
HPI_ldq_I0_m1		RealDisposablePersonalIncome_dq_I3_m2		const		NaN	NaN	0.566697
RealGDP_qoq_I2_m2		HPI_ldq_I0_m1		RealDisposablePersonalIncome_ldq_I2_m3		const	NaN	0.524809
RealDisposablePersonalIncome_yoy_I2_m2		LightWeightVehicleSales_lvl_I1_m1		const		NaN	NaN	0.47219

Model A: Top Performing



Key Observations

- Max total lag: **7Q** (YoY, 2Q lag, 2Q MA)
- Captures 2021 trough and 2023 peak
- 2011-2020 captured relatively well
- **Low test population adj r-squared**

Metrics	Full	Train	Test
Adj R2	70%	72%	24%
RMSE	0.0013	0.0012	0.0015

RootVariable	Transform	Lag	MovingAvg	Coef	P-value	VIF
HPI	ldQ	0	1	-0.0009	1.5%	1.66
LightWeightVehicleSales	lvl	1	1	0.0016	0.0%	1.67
RealDisposablePersonalIncome	YoY	2	2	-0.0012	0.0%	1.56
RealGDP	YoY	0	1	-0.0005	4.1%	1.41
Constant				0.0061	0.0%	

Model A: Variable Trends

Transformations

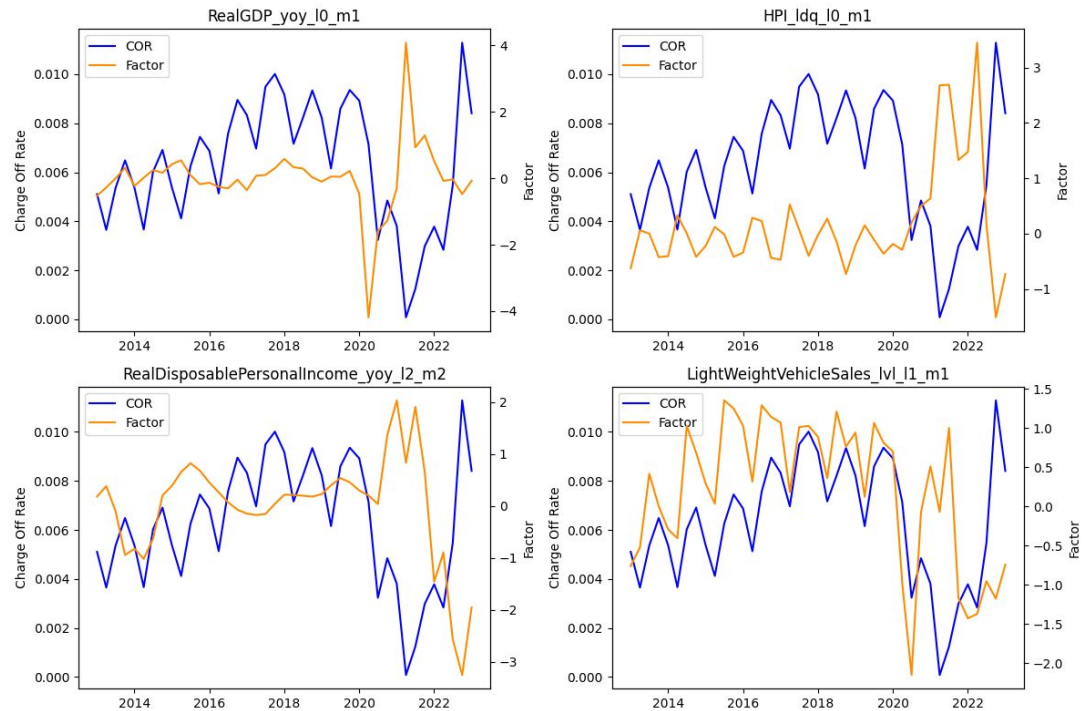
Two YoY transforms are applied to Real GDP and Real Disposable Personal Income.

Seasonality

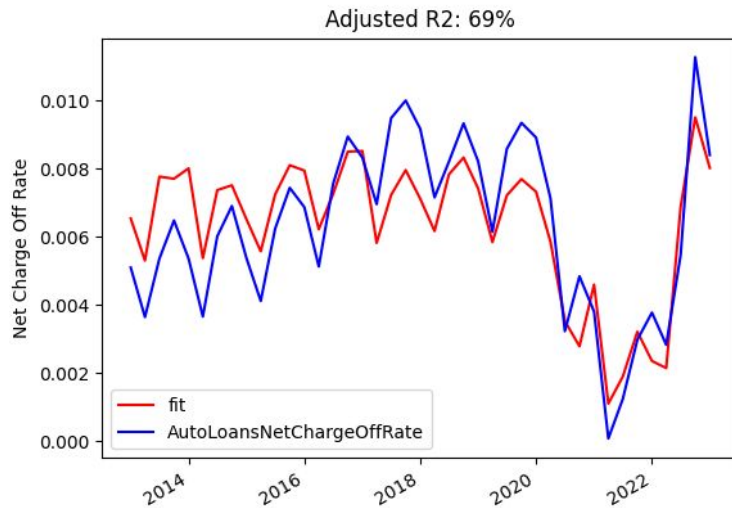
Seasonality is captured and represented by HPI and LWVS.

Key Trends

The 2021 trough and 2023 peak are directly attributed to all four model variables.



Model B: Reduced Lag



Key Observations

- Max total lag: **5Q** (no YoY variables)
- Captures 2021 trough and 2023 peak
- **Poorer test performance than A**
- 2011-2020 captured relatively well
- **Highest test population adj r-squared**

Metrics	Full	Train	Test
Adj R2	67%	68%	42%
RMSE	0.0013	0.0013	0.0014

RootVariable	Transform	Lag	MovingAvg	Coef	P-value	VIF
HPI	ldQ	0	1	-0.0018	0.0%	1.04
LightWeightVehicleSales	lvl	1	1	0.0011	0.3%	1.26
RealDisposablePersonalIncome	ldQ	2	3	-0.0009	0.2%	1.24
Constant				0.0064	0.0%	

Model B: Variable Trends

Transformations

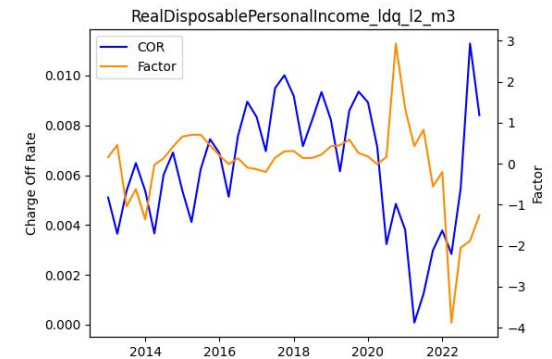
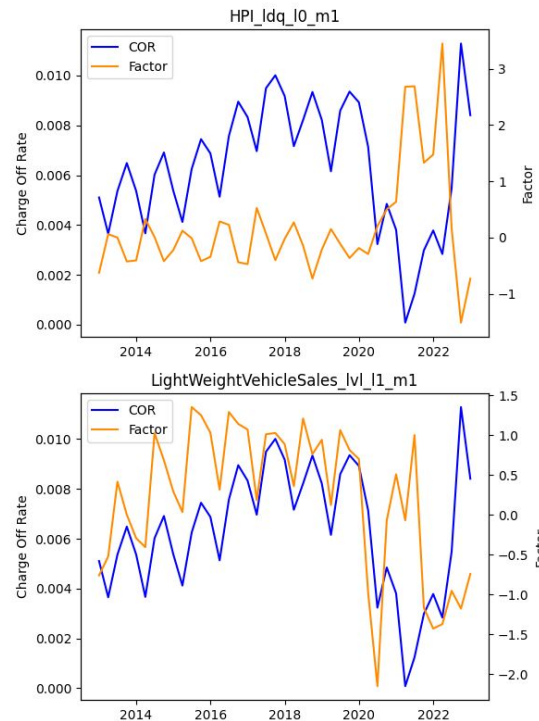
No YoY transforms are applied to the variables in Model B.

Seasonality

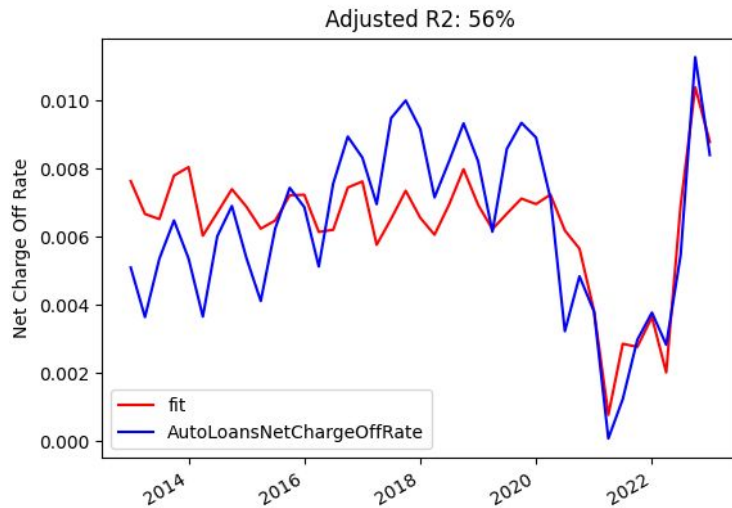
HPI and LWVS contribute significantly to seasonality.

Key Trends

The 2021 trough and 2023 peak are directly attributed to all four variables.



Model C: No LWVS



Key Observations

- Max total lag: 5Q
- Excluding LWVS
- 2021 trough and 2023 peak captured relatively well
- **Seasonality during 2011-2020 not well captured**
- Low test population adj r-squared

Metrics	Full	Train	Test
Adj R2	54%	58%	22%
RMSE	0.0016	0.0015	0.0018

RootVariable	Transform	Lag	MovingAvg	Coef	P-value	VIF
HPI	IdQ	0	1	-0.0019	0.0%	1.02
RealDisposablePersonalIncome	dQ	3	2	-0.0006	4.7%	1.02
Constant				0.0067	0.0%	

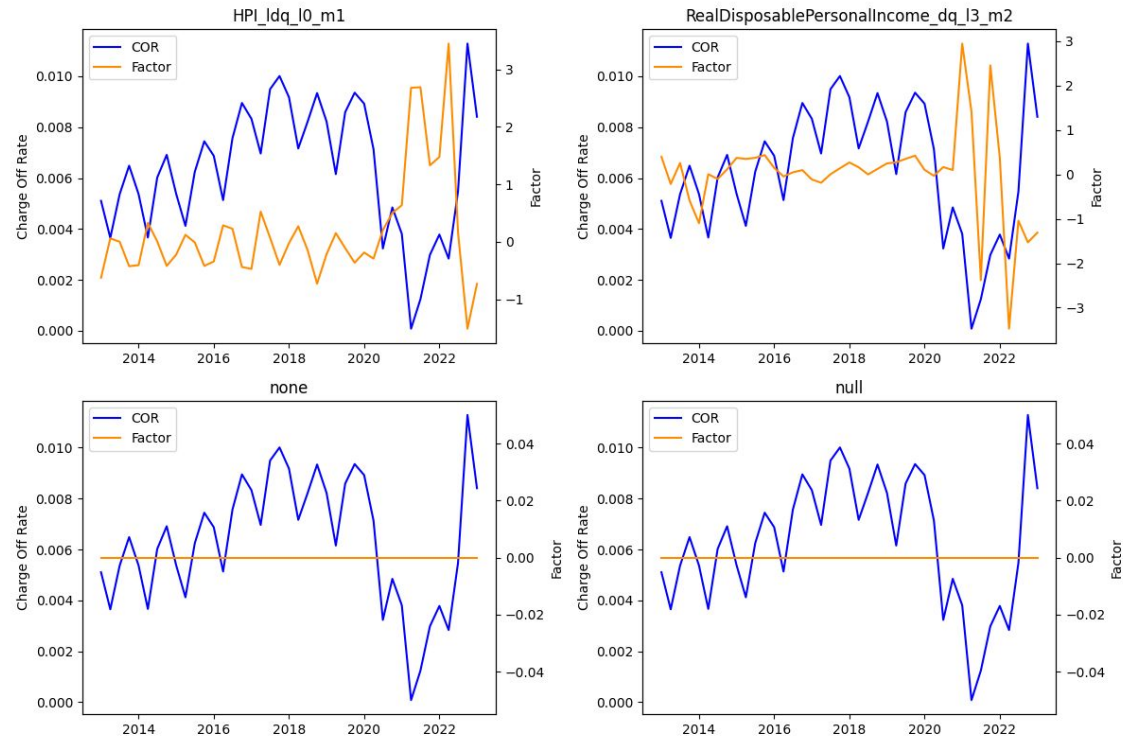
Model C: Variable Trends

Pre-2020 Trends

Pre-2020 trends are not captured well without the inclusion of Light-Weight Vehicle Sales (LWVS).

Seasonality

Seasonality is effectively captured and explained by the House Price Index (HPI) variable.



Model Recommendation

✓ Model B is recommended based on the model performance comparison below.

Metric	Model	Full	Train	Test
Adj R2	A	70%	72%	24%
	B	67%	68%	42%
	C	54%	58%	22%
RMSE	A	0.0013	0.0012	0.0015
	B	0.0013	0.0013	0.0014
	C	0.0016	0.0015	0.0018

